EXPERIMENT – 6

IMPLEMENTATION OF SLIDING WINDOW PROTOCOL

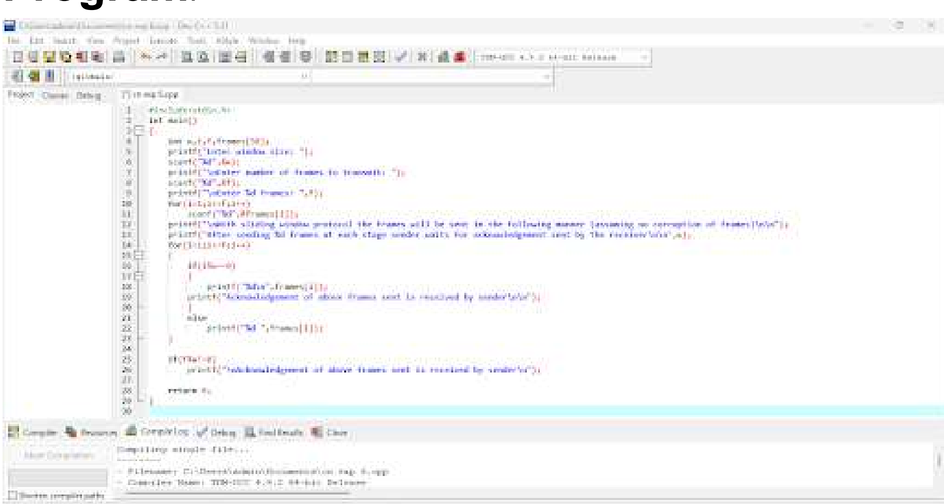
Aim

To implement the **Sliding Window Protocol** using a C/C++ program and simulate the transmission of frames between sender and receiver, verifying the acknowledgement mechanism.

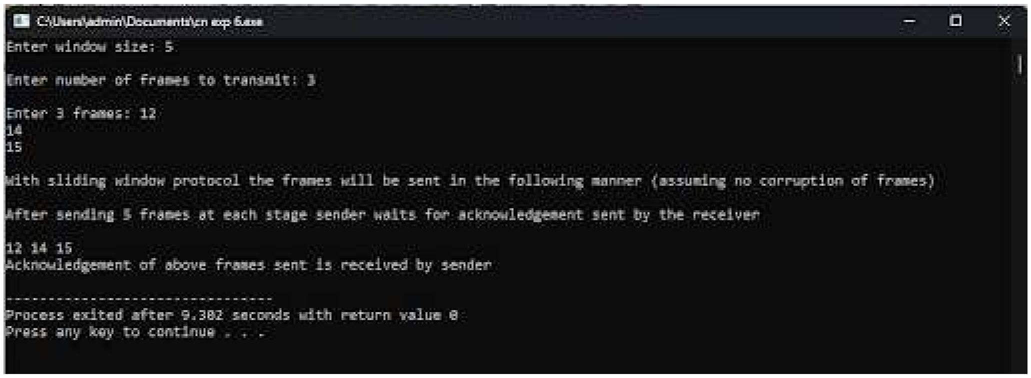
Procedure

1. Open **Dev C++ / any C compiler**.
2. Create a new **C/C++ program**.
3. Write the program to implement the **Sliding Window Protocol**.
4. The program accepts:
    - Window size
    - Number of frames to be transmitted
5. Enter the frames that need to be sent.
6. The sender sends frames according to the **window size**.
7. After sending frames within the window, the sender waits for **acknowledgement from the receiver**.
8. If acknowledgement is received, the window **slides forward** to send the next frames.
9. Compile and run the program.
10. Observe the frame transmission and acknowledgement messages in the output.

Program:



Output:



Observation

1. The sliding window protocol allows multiple frames to be sent before waiting for acknowledgement.
2. Frames are transmitted according to the specified window size.
3. After sending frames within the window, the sender waits for acknowledgement from the receiver
4. Once acknowledgement is received, the sender proceeds to send the next set of frames.
5. This method improves transmission efficiency compared to stop-and-wait protocol.

Result

The Sliding Window Protocol was successfully implemented using a C/C++ program, and the transmission of frames along with acknowledgement handling was verified through the program output.